Lemma F for the octahedron: opposite petals never overlap

working notes

September 2026 — draft

Setting and conventions

We use the conventions of `HANDOFF.md`. P is a convex polytope of octahedron type with antipodal (non-adjacent) vertices v, w and equator $u_0, \dots, u_3$ (the neighbours of w, indices mod 4, in counter-clockwise order around w seen from outside). The faces are $W_i = wu_iu_{i+1}$ and $V_i = vu_iu_{i+1}$; κ_x is the curvature at x; ω_i, ν_i are the angles of W_i at w and of V_i at v. The net Z_k cuts $\mathrm{star}(v) \cup \{wu_k\}$: the fan $W_k, W_{k+1}, W_{k+2}, W'_{k-1}$ around w (angular gap κ_w at the slit) with the petal V_i glued to W_i along u_iu_{i+1}. Touching counts as non-overlapping.

Only nine face pairs of Z_k share no net vertex: the three local pairs at the slit, the two opposite-petal pairs (V_k, V_{k+2}), (V_{k+1}, V'_{k-1}), and four petal-vs-far-fan pairs. Lemma F is the statement that under the hypothesis

$$\textbf{(H)} \qquad \kappa_v \;\geq\; \kappa_x \quad \text{for every vertex } x \text{ of } P \tag{1}$$

none of the six far pairs overlaps, in every Z_k. (The handoff's hypothesis $\kappa_v \geq \max_i \kappa_{u_i}$ is weaker; the numerics below show that the weaker hypothesis is *not* enough for the argument of §3, although no counterexample to the weaker statement itself is known.)

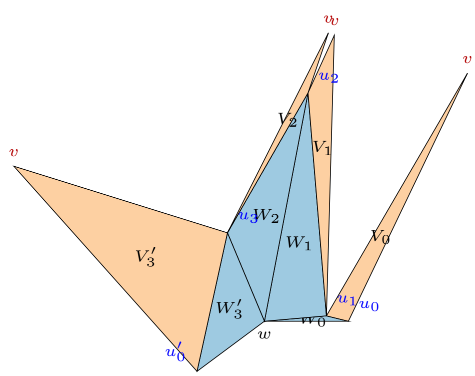


Figure 1: A typical net Z_k (fan faces blue, petals orange), v the sharpest vertex, slit at the sharpest neighbour u_0 of w.

Elementary consequences of (H). Since $\sum_x \kappa_x = 4\pi$ over six vertices, (H) gives $\kappa_v \geq 2\pi/3$, hence $\sum_i \nu_i = 2\pi - \kappa_v \leq 4\pi/3$. At any convex vertex each face angle is at most the sum of the other face angles (the spherical polygon inequality for the tangent cone), so $\nu_j \leq \frac{1}{2}\sum_i \nu_i$, i.e.

$$\nu_j \;\leq\; \pi - \tfrac{1}{2}\kappa_v \;\leq\; \tfrac{2\pi}{3} \qquad \text{for every } j. \tag{2}$$

1 Reduction to three consecutive petals

Lemma 1 (petals do not cross interior hinges)**.** *In Z_k no petal meets the relative interior of a hinge segment wu_j, $j \neq k$.*

Proof. The two fan faces adjacent to wu_j are W_{j-1} and W_j; a petal crossing the segment would enter both. Every petal shares a net vertex with W_{j-1} or W_j (V_{j-1}, V_j share u_j; V_{j+1} shares u_{j+1} with W_j; V_{j-2} shares u_{j-1} with W_{j-1}), and faces sharing a net vertex lie in disjoint wedges there. □

Lemma 2 (petal versus far fan face)**.** *If a petal V_i overlaps a fan face W_j with $j \notin \{i-1, i, i+1\}$ then V_i overlaps V_j.*

Proof. V_i contains no vertex of W_j (w and the equator vertices of W_j lie in fan faces or petals that share a vertex with V_i, except possibly the local face W'_{k-1}, whose treatment is part of Lemma L). By Lemma 1 the boundary of V_i cannot enter W_j through its hinges, so it enters through the outer edge $u_j u_{j+1}$, which is the common edge of W_j and V_j; near that crossing V_i meets V_j. □

Hence Lemma F follows from the following statement about three consecutive petals, applied to the triples (V_k, V_{k+1}, V_{k+2}) and $(V_{k+1}, V_{k+2}, V'_{k-1})$ of Z_k.

Conjecture 3 (Triple Lemma)**.** *Let W_i, W_j, W_{j+1} ($j = i+1$) be three consecutive fan faces of a net and V_i, V_j, V_{j+1} their petals. Under (H), V_i and V_{j+1} do not overlap.*

Numerical evidence 4. No overlap of opposite petals under (H) in 72,000 random nets, nor under the weaker hypothesis $\kappa_v \geq \max \kappa_u$; adversarial hill-climbing (edges ≥ 0.05 diam, curvatures ≥ 0.05) brings the two petals to within 0.4% of the diameter but never into overlap. All observed overlaps (44 in 72,000 nets, without hypothesis) have two equator vertices sharper than v.

2 Where a meeting can be

Fix a triple as in Conjecture 3 and write $u = u_j$, $u' = u_{j+1}$ for the base of V_j, ϕ, ψ for the base angles of V_j at u, u', $\kappa = \kappa_u + \kappa_{u'}$. In the net, V_i and V_j meet at u only, separated there by the angular gap κ_u between the two copies uv_j and uv_i of the cut edge uv; likewise at u'. Let L be the line through uv_i and L' the line through $u'v_{j+1}$.

Lemma 5 (double-outward wedge)**.** *V_i lies in the closed half-plane of L not containing V_j, and V_{j+1} in the closed half-plane of L' not containing V_j. Hence every common point of V_i and V_{j+1} lies in $\mathcal{D}$:= the intersection of these two half-planes, which is the wedge at the point $p^* = L \cap L'$ opposite to the base uu'. If $\kappa < \nu_j$ then $\mathcal{D}$ lies on v_j's side of the base line (above), and if $\kappa > \nu_j$ it lies on w's side (below); $\mathcal{D} = \emptyset$ if $\kappa = \nu_j$.*

Proof. The rays uv_i and $u'v_{j+1}$ make angles $\phi + \kappa_u$ and $\psi + \kappa_{u'}$ with the base, so they converge above the base iff $\phi + \psi + \kappa < \pi$, i.e. $\kappa < \nu_j$; the rest is the description of the four wedges at p^*. □

Lemma 6 (the two cut edges never cross)**.** *The segments uv_i and $u'v_{j+1}$ are disjoint.*

Proof. If they crossed at p then p would be above the base with base angles larger than ϕ, ψ, so v_j lies inside the triangle $uu'p$ and $|up| + |u'p| > |uv_j| + |u'v_j| = |uv_i| + |u'v_{j+1}|$, contradicting $|up| \leq |uv_i|$, $|u'p| \leq |u'v_{j+1}|$. □

Claim 7 (observed overlaps)**.** *In the 44 observed opposite-petal overlaps (no hypothesis), the intersection region lies in $\mathcal{D}$; 26 are above the base and 18 below, exactly as predicted by the sign of $\nu_j - \kappa$. In every below-base case the outer edges uu_i and $u'u_{j+2}$ of W_i, W_{j+1} converge on w's side (equivalently $\angle_u W_i + \angle_u W_j + \angle_{u'} W_j + \angle_{u'} W_{j+1} < \pi$) and exactly one of the two fan faces extends past the crossing point; both cannot, since fan faces are disjoint.*

3 The rotation picture (above-base case)

Glue V_i, V_j, V_{j+1} flat around v with no gaps (the "v-fan"); this is possible because $\nu_i+\nu_j+\nu_{j+1} < 2\pi$, and in it the three petals occupy disjoint wedges at v. The net position of V_i is obtained from its v-fan position by the rotation ρ_u about u by the gap angle κ_u (away from V_j), and that of V_{j+1} by the rotation $\rho_{u'}$ about u' by $\kappa_{u'}$. Both rotations have the same sense. Therefore

$$V_i \cap V_{j+1} \neq \emptyset \text{ in the net} \iff \mathcal{R}(V_i^{\text{fan}}) \cap V_{j+1}^{\text{fan}} \neq \emptyset, \qquad \mathcal{R} := \rho_{u'}^{-1} \circ \rho_u,$$

and $\mathcal{R}$ is the rotation by $\kappa = \kappa_u + \kappa_{u'}$ about the point c^* on w's side of the base with $\angle c^* uu' = \kappa_u/2$, $\angle uu'c^* = \kappa_{u'}/2$ (composition of two rotations). Let M be the line c^*v; $\mathcal{R}$ moves points from u''s side of M towards u's side. Measure angles around c^* from M, positively towards u's side.

Lemma 8 (angular criterion)**.** *Let $\varepsilon_i \geq 0$ be the largest angle (seen from c^*) by which $V_i^{\text{fan}} \cap \{$above the base$\}$ crosses M towards u''s side, and ε_{j+1} the corresponding excursion of $V_{j+1}^{\text{fan}} \cap \{$above$\}$ towards u's side. If $\kappa < \nu_j$ and*

$$\varepsilon_i + \varepsilon_{j+1} \ < \ \kappa, \tag{3}$$

then V_i and V_{j+1} do not overlap in the net.

Proof. By Lemma 5 a common point p is above the base. The rotation ρ_u^{-1} maps the wedge of V_i at u in the net, $[\phi + \kappa_u,\ 2\pi - \angle_u W_j - \angle_u W_i]$, onto the v-fan wedge, so $x = \rho_u^{-1}(p)$ is below the base only if p is; hence $x \in V_i^{\text{fan}}$ and $y = \rho_{u'}(p) \in V_{j+1}^{\text{fan}}$ are above the base and $y = \mathcal{R}(x)$. Their angles from M satisfy $\theta_y = \theta_x + \kappa$ with $\theta_x \geq -\varepsilon_i$ and $\theta_y \leq \varepsilon_{j+1}$, so $\kappa \leq \varepsilon_i + \varepsilon_{j+1}$ (no wrap-around since $\kappa < \nu_j < \pi$). $\square$

In particular, if neither petal crosses M above the base ($\varepsilon_i = \varepsilon_{j+1} = 0$), there is no overlap. When does a petal cross M? Write $\mu_u = \angle uvc^*$, $\mu_{u'} = \angle c^* vu'$, so $\mu_u + \mu_{u'} = \nu_j$ (c^* lies in the cone of V_j at v, being on w's side of the base between the lines vu, vu'). The wedge of V_i at v starts at vu and turns away from V_j by ν_i; it reaches the ray of M beyond v iff $\nu_i > \pi - \mu_u$. The other way for $V_i^{\text{fan}} \cap \{\text{above}\}$ to cross M is through the far vertex u_i or through the point where the edge vu_i or uu_i crosses the base line.

Lemma 9. *Under (H) the two petals cannot both cross M through their wedges at v.*

Proof. That would need $\nu_i + \mu_u > \pi$ and $\nu_{j+1} + \mu_{u'} > \pi$, hence $\nu_i + \nu_{j+1} + \nu_j > 2\pi$, whereas $\nu_i + \nu_j + \nu_{j+1} \leq \sum \nu \leq 4\pi/3$. $\square$

Lemma 10 (the above-base case needs a very sharp v)**.** *Under (H), $\kappa < \nu_j$ forces $\kappa_v > 6\pi/7$, hence $\sum_i \nu_i < 8\pi/7$ and each $\nu_i < 4\pi/7$.*

Proof. $4\pi = \sum_x \kappa_x \leq 4\kappa_v + \kappa$ and $\kappa < \nu_j \leq \pi - \kappa_v/2$ by (2). $\square$

Numerical evidence 11. In 3000 random octahedra (all apexes, all triples with $\kappa < \nu_j$): the composition $\mathcal{R}$ has the stated centre and sense, and the criterion of Lemma 8 reproduces the net overlap test exactly. Under (H) neither petal ever crosses M above the base ($\varepsilon_i = \varepsilon_{j+1} = 0$ in all 224 triples, degenerate ones included), and adversarial search under (H) cannot make either excursion positive. Under the weaker hypothesis $\kappa_v \geq \max \kappa_u$ a petal can cross M and (3) can fail (excursion 1.96 with $\kappa = 0.05$), so (H) is genuinely used. Every one of the 26 observed above-base overlaps violates $\varepsilon_i = \varepsilon_{j+1} = 0$ (20 through V_i, 6 through V_{j+1}).

Superseded: what remained for the rotation argument. Prove that under (H) and $\kappa < \nu_j$ neither $V_i^{\text{fan}} \cap \{\text{above}\}$ nor $V_{j+1}^{\text{fan}} \cap \{\text{above}\}$ crosses M (or, more weakly, that (3) holds). The three crossing mechanisms are: (a) the wedge at v: $\nu_i > \pi - \mu_u$; (b) the far vertex u_i above the base on u''s side of M; (c) an edge of V_i crossing the base line outside the segment uu' on u''s side of M. Mechanism (c) through vu_i requires $\nu_i + \nu_j > \pi + \psi$; (a) requires $\nu_i + \nu_j > \pi$; by Lemma 10 these need $\kappa_v < \pi$ and two of the four angles at v close to $4\pi/7$. The numerics say the fan faces then obstruct this, but the argument is not yet written.

4 Case A is closed: apex cones

The side conditions of §3 are not needed. Two lemmas finish the above-base case.

Lemma 12 (no wrap). *Under (H), $\nu_i + \nu_j \leq \pi + \kappa_u + \kappa_{u'}$ and $\nu_{j+1} + \nu_j \leq \pi + \kappa_u + \kappa_{u'}$ for every triple.*

Proof. From $\sum_x \kappa_x = 4\pi$ and (H), $\kappa := \kappa_u + \kappa_{u'} = 4\pi - \kappa_v - \kappa_w - \kappa_{u_i} - \kappa_{u_m} \geq 4\pi - 4\kappa_v = 4\sum \nu - 4\pi$. If $\sum \nu \geq \pi$ then $\pi + \kappa \geq 4\sum \nu - 3\pi \geq \sum \nu \geq \nu_i + \nu_j$; if $\sum \nu < \pi$ then $\nu_i + \nu_j < \pi < \pi + \kappa$. $\square$

Lemma 13 (apex cones). *Let C_i be the closed cone at v_i spanned by the cut edges $v_i u$ and $v_i u_i$ (it contains V_i), and C_{j+1} the cone at v_{j+1} spanned by $v_{j+1}u'$ and $v_{j+1}u'_m$. If $\kappa < \nu_j$, $\nu_i \leq \pi - \nu_j + \kappa$ and $\nu_{j+1} \leq \pi - \nu_j + \kappa$, then C_i and C_{j+1} have disjoint interiors. In particular V_i and V_{j+1} do not overlap.*

Proof. Measure directions counter-clockwise from $u \to u'$ and put $\alpha = \phi + \kappa_u$, $\beta = \psi + \kappa_{u'}$, so $\alpha + \beta = \pi - \nu_j + \kappa < \pi$. The cut edge uv_i has direction α (the gap turns v_j away from V_j) and V_i lies on its left, so from v_i the cone C_i is the sector $I_i = [\pi + \alpha - \nu_i,\ \pi + \alpha]$ ending at the direction e_1 towards u. Likewise $u'v_{j+1}$ has direction $\pi - \beta$, V_{j+1} lies on its right, and C_{j+1} is the sector $I_{j+1} = [-\beta,\ -\beta + \nu_{j+1}]$ starting at the direction towards u'. Two cones meet iff $d := v_{j+1} - v_i$ lies in the convex cone K generated by $I_i \cup (-I_{j+1})$, where $-I_{j+1} = [\pi - \beta,\ \pi - \beta + \nu_{j+1}]$. The two hypotheses say exactly that I_i and $-I_{j+1}$ lie inside $S = [\pi - \beta,\ \pi + \alpha]$, a sector of width $\alpha + \beta < \pi$ bounded by e_1 and by $f_2 =$ the direction of $u' \to v_{j+1}$; hence $K = S$. Now $d = s_u e_1 + \ell e_0 + s_{u'} f_2$ with e_0 the base direction, and writing $e_0 = c_1 e_1 + c_2 f_2$ gives $c_1 = -\sin\beta / \sin(\alpha + \beta)$, $c_2 = -\sin\alpha / \sin(\alpha + \beta)$. Thus $d = (s_u - |up^*|)e_1 + (s_{u'} - |u'p^*|)f_2$ where p^* is the crossing of the two cut-edge lines, and $d \in S$ iff $s_u \geq |up^*|$ and $s_{u'} \geq |u'p^*|$, i.e. iff both cut edges reach p^*, i.e. iff they cross. Lemma 6 excludes this (equality means the cones touch at p^*). $\square$

Proposition 14. *Under (H) the Triple Lemma holds whenever $\kappa_u + \kappa_{u'} < \nu_j$ (Case A).*

Proof. Lemma 12 gives the two angle conditions of Lemma 13. $\square$

Numerical evidence 15. On 3241 random Case-A triples without hypothesis the cones never meet when the two angle conditions hold, and the conditions fail in 8% of them; under (H) they never fail (208 triples; adversarial margin 1.3 rad, `adv_wrap.py`), and the cone intersection has empty interior in every Case-A triple (adversarial depth -0.0007 diameters at the curvature guard, `adv_cones.py`). Without hypothesis the adversary makes the cones meet (depth $+0.006$), as it must, since 26 above-base overlaps exist.

Case B in this language. Any meeting lies in $C_i \cap C_{j+1}$; in Case B this intersection is often non-empty (15% of Case-B triples under (H) have both petals reaching it), so the cones alone do not suffice. The tight Case-B configurations are meetings around the fourth petal V_m at the slit; the D-lemma applied there shows the meeting must lie beyond the crossing X of the outer-edge lines of W_i and W_{j+1} (which requires the four fan angles at u, u' to sum to less than π), and adversarially the petals stay 3.4% of the diameter apart inside that wedge under (H). Open.

5 The below-base case

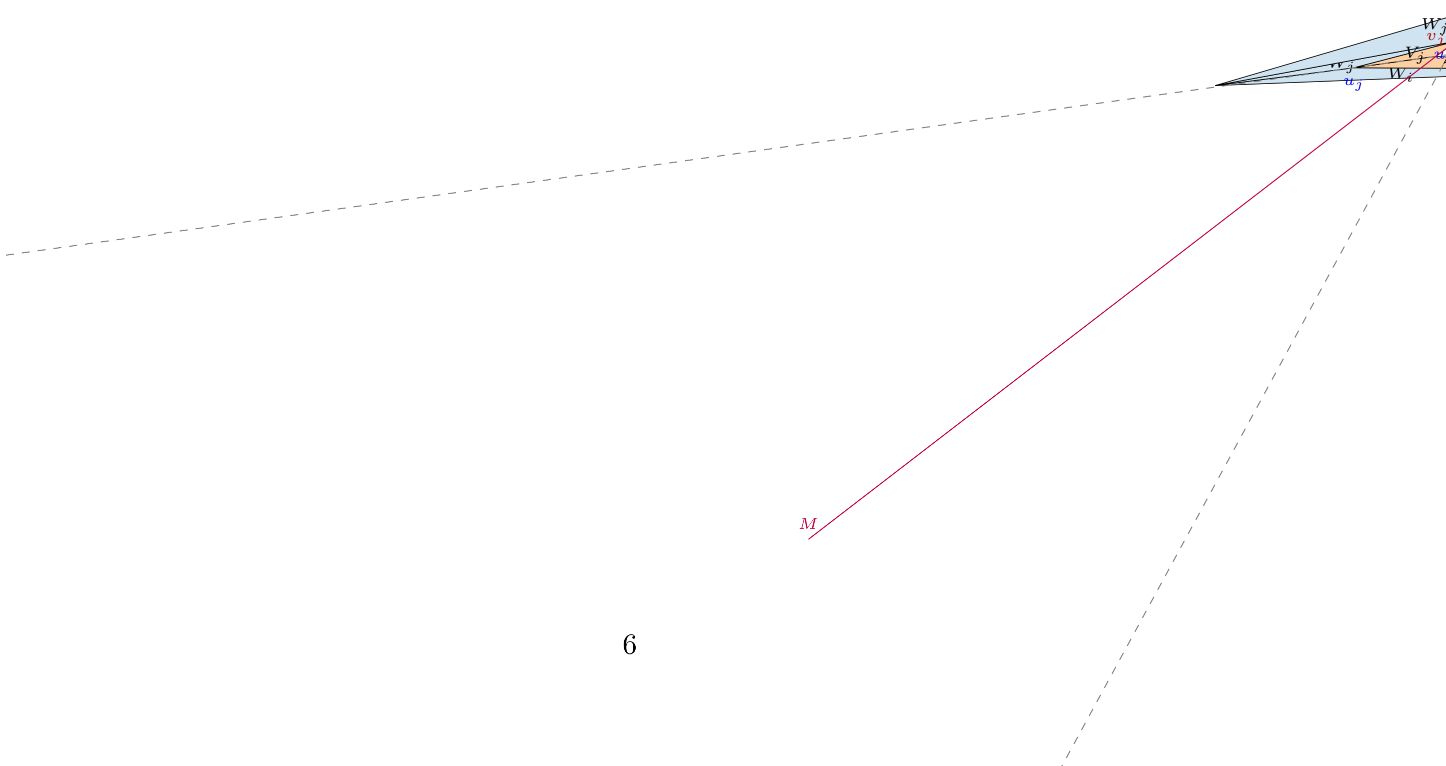
When $\kappa > \nu_j$ the wedge $\mathcal{D}$ lies on w's side of the base. Then (Claim above) a meeting requires the outer-edge lines of W_i and W_{j+1} to converge on w's side, one of the two fan faces to reach past their crossing point X, and the opposite petal to pass over the far vertex of that fan face. In the observed below-base overlaps two equator vertices have curvature close to 2π and v is nearly flat. Under (H), $\kappa_u, \kappa_{u'} \leq \kappa_v$ and $\nu_j < \kappa \leq 2\kappa_v$; no proof yet.

6 Known false / do not retry

Simple separating lines for the two petals inside $\mathcal{D}$ (bisector of $\mathcal{D}$ at p^*, lines through v_j) fail in about a third of the non-degenerate cases even under (H): the separation is a rotation phenomenon, not a half-plane one. The six wedge–lean inequalities of the handoff fail in 0.5% of nets under (H) (needle-like w), so they cannot replace the argument above.

Code

`octa.py` (module: all quantities, nets Z_k, nine pairs, intersection polygons), `vwedge.py`, `oppcases.py` (collect/classify/draw overlaps), `adv_opp.py` (adversarial separation), `cstar.py` (checks of §3), `adv_eps.py`, `adv_mu.py`, `adv_cstar.py` (adversarial tests of (3) and of the crossing conditions), `tikz_net.py` (figures), `draw.py` (PNG).


W_j
v_j
V_j
w_j
u_j
W_i
M

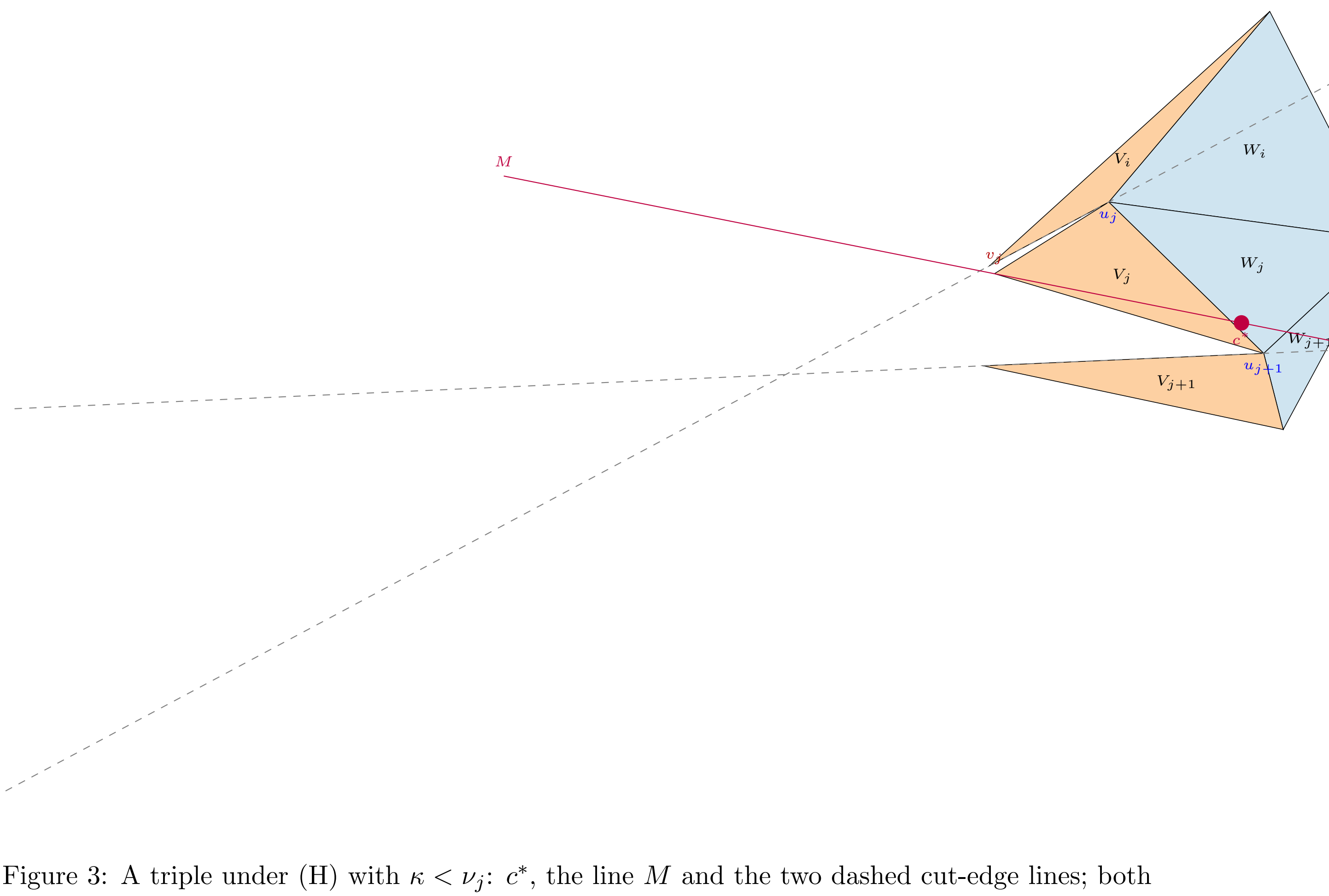


Figure 3: A triple under (H) with $\kappa < \nu_j$: c^*, the line M and the two dashed cut-edge lines; both petals stay on their own side of M.